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Atherosclerosis: a reaction–diffusion wave

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Summary

1 Introduction.

2 Atherogenesis mechanism.

3 Mono-dimensional inflammatory model

4 Numerical simulations.

5 Conclusion and perspectives

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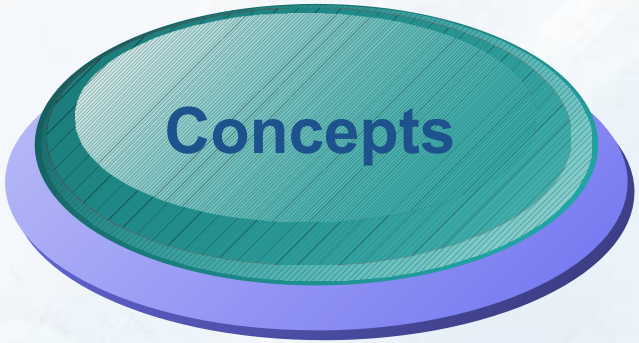
Introduction



Arteriosclerosis: Hardening of the arteries.

Atherosclerosis deposition of lipids, not pasive, inflamatory.

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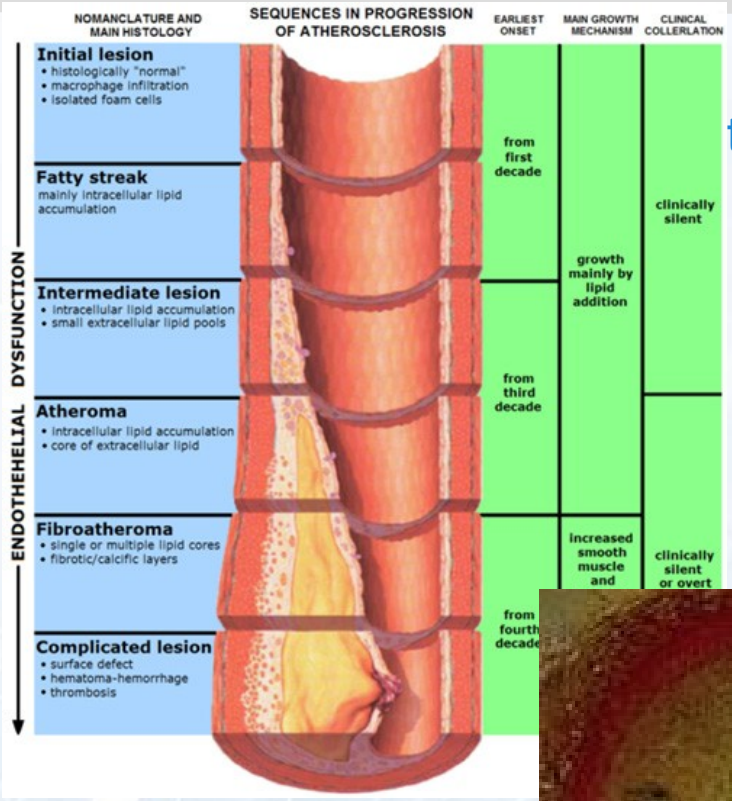
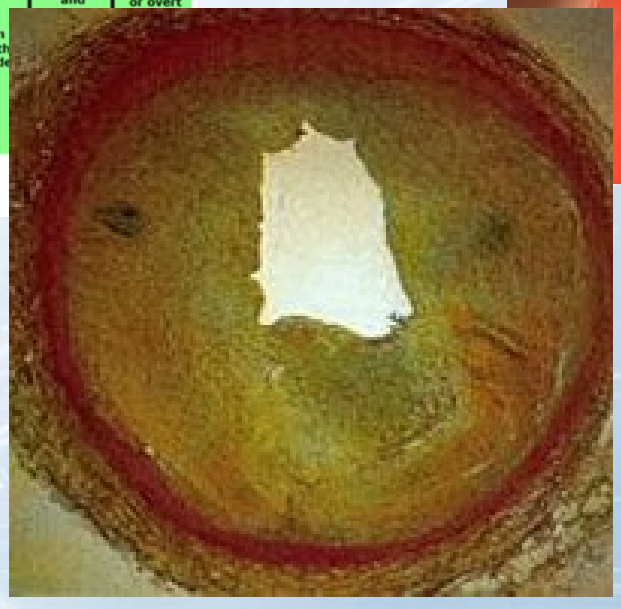
Atherogenesis mechanism

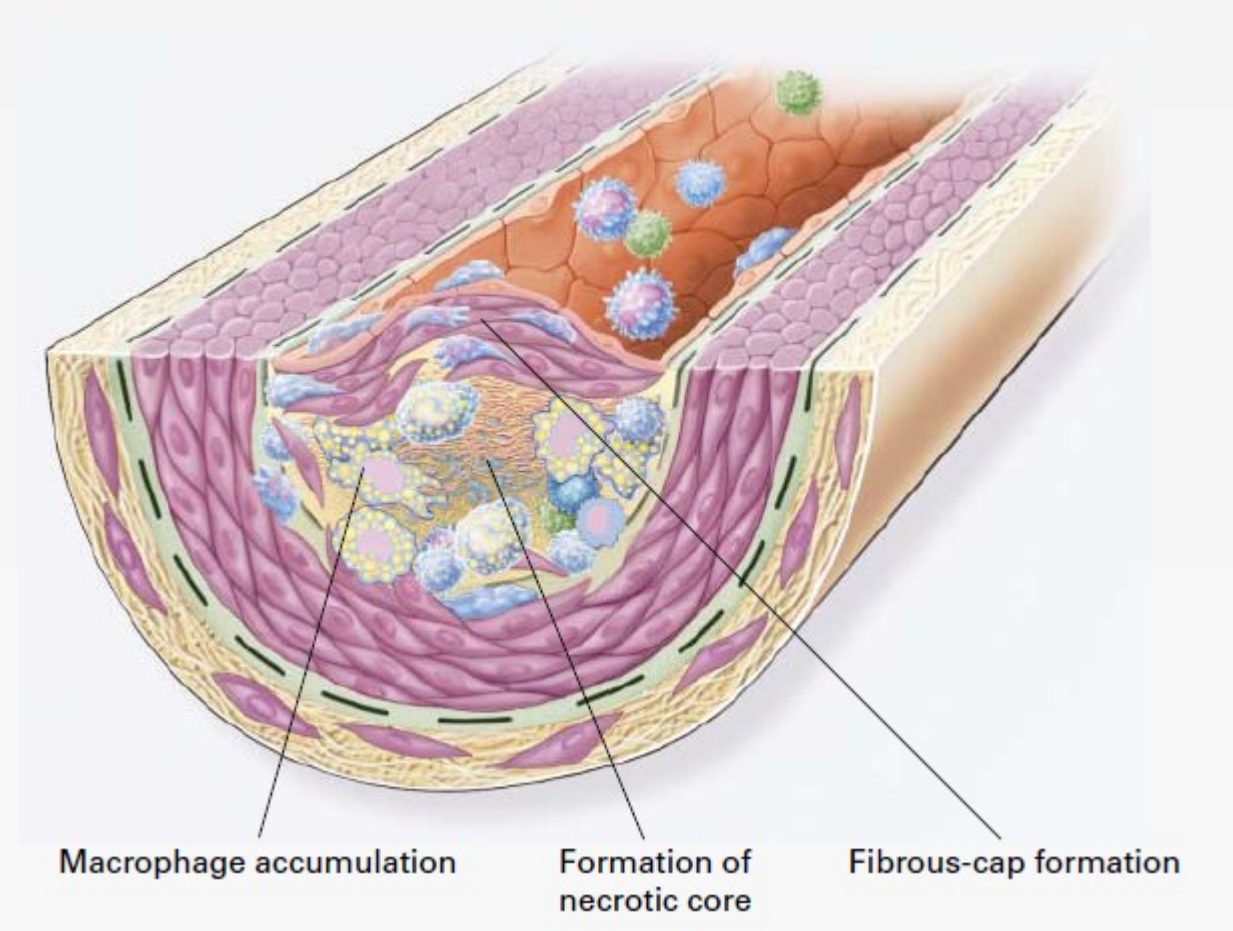


evolution of the disease to the complete formation of the fibrous cape



Transverse cut of an partially obstructed artery

Atherogenesis mechanism



R Ross. Atherosclerosis. An inflammatory disease, *N Engl J Med* 340 (1999), pp. 115-126.

Atherogenesis mechanism

- The process begins when the LDLs penetrate into the intima of the arterial wall where they are oxidized.
- The immune system considers this LDL-ox like a dangerous agent and it provokes an immune answer, which mediates the adhesion of the monocytes to the endothelium.
- Once there they are converted in macrophages.
- Then appears pro-inflammatory cytokines

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Mono-dimensional inflammatory model

$$\begin{cases} \dfrac{\partial M}{\partial t} = d_1 \dfrac{\partial^2 M}{\partial x^2} + f_1(A) - \lambda_1 M, \\[2ex] \dfrac{\partial A}{\partial t} = d_2 \dfrac{\partial^2 A}{\partial x^2} + f_2(A) M - \lambda_2 A, \end{cases}$$

M: Monocytes
A: Inflammatory Cytokines

$$f_1(A) = \frac{\alpha_1 + \beta_1 A}{1 + A/\tau_1} \qquad f_2(A) = \frac{\alpha_2 A}{1 + A/\tau_2}$$

$f_1(A)$: recruitment of the immune cells from the blood flow
$f_2(A)$: cytokines production rate

1-D Model

1-D Model

- α_2 corresponds to the secretion of cytokines

- $1 + \frac{A}{\tau_2}$ represents the inhibition of this secretion, being τ_2 the time for it

System without diffusion

Blood Diffusion

$$\begin{cases} \dfrac{dM}{dt} = f_1(A) - \lambda_1 M \\ \dfrac{dA}{dt} = f_2(A)M - \lambda_2 A \end{cases} \Rightarrow \begin{cases} F_1(A) = M = \dfrac{f_1(A)}{\lambda_1} \\ F_2(A) = M = \dfrac{\lambda_2 A}{f_2(A)} \end{cases}$$

Equilibrium Points

$$\begin{cases} \dfrac{dM}{dt} = f_1(A) - \lambda_1 M \\ \dfrac{dA}{dt} = f_2(A)M - \lambda_2 A \end{cases}$$

$$E_0 = \left(\frac{\alpha_1}{\lambda_1}, 0\right)$$

E_r y roots of equation

$$F_1(A) = F_2(A)$$

$$\frac{\lambda_2\lambda_2}{\alpha_2\tau_1\tau_2}A^2 + \left(\frac{\lambda_2\lambda_2}{\alpha_2\alpha_2}\left(\frac{1}{\tau_1}+\frac{1}{\tau_2}\right)\frac{1}{\tau_2}\right)\frac{\beta_1\beta_1}{\lambda_1\lambda_1}A + \frac{\lambda_2\lambda_2}{\alpha_2\alpha_2}\frac{\alpha_1\alpha_1}{\lambda_1\lambda_1} = 0$$

Equilibrium Points

Denote p y s to the product and sum of roots of :

$$\frac{\lambda_2\lambda_2}{\alpha_2\tau_1\tau_2}A^2 + \left(\frac{\lambda_2\lambda_2}{\alpha_2}\left(\frac{1}{\tau_1}+\frac{1}{\tau_2}\right)-\frac{\beta_1\beta_1}{\lambda_1\lambda_1}\right)A + \frac{\lambda_2\lambda_2}{\alpha_2}\frac{\alpha_1\alpha_1}{\lambda_1\lambda_1} = 0$$

Then:

$$p = \left(\alpha - \frac{\alpha_1}{\lambda_1}\right)\frac{\tau_1}{\beta}$$

$$s = \left(\frac{\beta_1}{\lambda_1} - \frac{\alpha}{\tau_1} - \beta\right)\frac{\tau_1}{\beta}$$

Equilibrium Points

- Case 1:

then only one positive root:

$\frac{\lambda_2}{\alpha_2} < \frac{\gamma_1}{\lambda_1}$ then $p < 0 \Rightarrow$ only one positive root: A_r

EquilibriumPoint:

Equilibrium Point: $E_r = (M_r, A_r)$

Equilibrium Points

- Case 2:

then $\frac{\alpha_1}{\lambda_1} < \frac{\lambda_2}{\alpha_2} < \frac{\beta_1 \tau_1}{\lambda_1}$ then $p > 0$:

- Case 2.1

then

- Case 2.1

Two positive solutions: $0 < \frac{\lambda_2}{\alpha_2 \tau_2} < \left(\sqrt{\frac{\beta_1}{\lambda_1} - \frac{\alpha_1}{\tau_1 \lambda_1}} - \sqrt{\frac{\lambda_2}{\alpha_2 \tau_1} - \frac{\alpha_1}{\tau_1 \lambda_1}}\right)^2$ then $s > 0$

Equilibrium Points: y

Two positive solutions: A_l y A_r

Equilibrium Points: $E_l = (M_l, A_l)$ y $E_r = (M_r, A_r)$

Equilibrium Point

- Case 2:

then $\frac{\alpha_1}{\lambda_1} < \frac{\lambda_2}{\alpha_2} < \frac{\beta_1 \tau_1}{\lambda_1}$ then $p > 0$:

- Case 2.2

then

- Case 2.2

$$0 < \left(\sqrt{\frac{\beta_1}{\lambda_1} - \frac{\alpha_1}{\tau_1 \lambda_1}} - \sqrt{\frac{\lambda_2}{\alpha_2 \tau_1} - \frac{\alpha_1}{\tau_1 \lambda_1}}\right)^2 < \frac{\lambda_2}{\alpha_2 \tau_2} \text{ then } s < 0$$

no positive solutions

There isn´t Equilibrium Points

no positive solutions

There isn´t Equilibrium Points

Equilibrium Point

- Case 3:
- Case 3:

$\frac{\beta_1 \tau_1}{\lambda_1} < \frac{\lambda_2}{\alpha_2}$ then no positive solutions

then $s < 0 \Rightarrow$ no positive solutions

Thereisn´tEquilibriumPoints

There isn´t Equilibrium Points

Stability

Stability

1.

$$J_{E_0} = \begin{pmatrix} -\lambda_1 & f_1'(0) \\ 0 & \alpha_2 \frac{\alpha_1}{\lambda_1} - \lambda_2 \end{pmatrix}$$

Eigenvalues: $\mu_1 = -\lambda_1, \quad \mu_2 = \frac{\alpha_1 \alpha_2}{\lambda_1} - \lambda_2$

If $\frac{\alpha_1}{\lambda_1} < \frac{\lambda_2}{\alpha_2} \Rightarrow$ $\boldsymbol{E_0}$ **stable**,

Otherwise **unstable.**

Otherwise $\Rightarrow$ $\boldsymbol{E_0}$ **unstable.**

Stability

2. $$J_{E_l} = \begin{pmatrix} -\lambda_1 & f_1'(A_l) \\ f_2(A_l) & f_2'(A_l) - \lambda_2 \end{pmatrix}$$

Like E_0 is concave and is a straight line:

$$F_1'(A_l) > F_2'(A_l) \Rightarrow \det(J_{E_l}) < 0$$

E_l is always unstable.

Stability

3. $$J_{E_r} = \begin{pmatrix} -\lambda_1 & f_1'(A_r) \\ f_2(A_r) & f_2'(A_r)\frac{\lambda_2 A_r}{f_2(A_r)} - \lambda_2 \end{pmatrix}$$

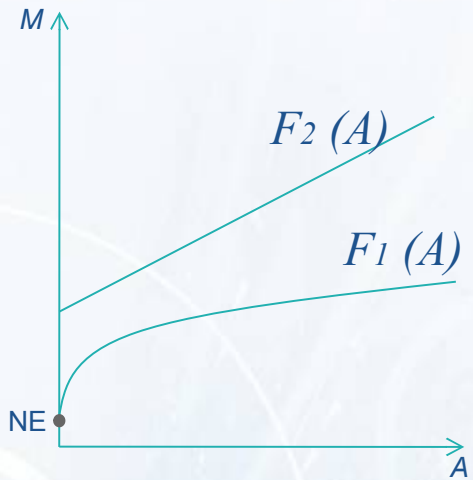
$$F_1'(A_r) < F_2'(A_r) \Rightarrow det(J_{E_r}) > 0$$

$$\Downarrow$$

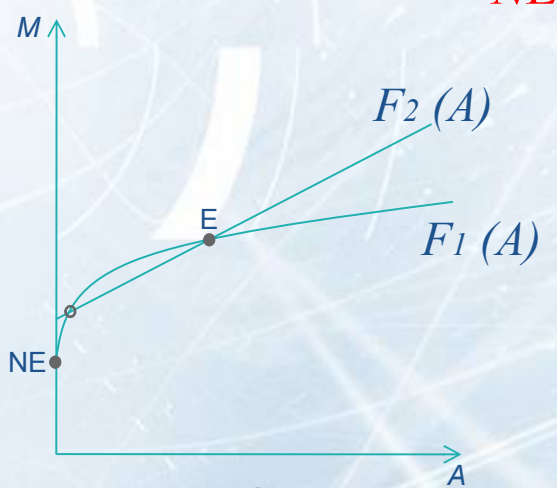
$$tr(J_{E_r}) = -\lambda_1 - \lambda_2 + f_2'(A_r)\frac{\lambda_2 A_r}{f_2(A_r)} < 0$$

E_r is always stable.

Stability



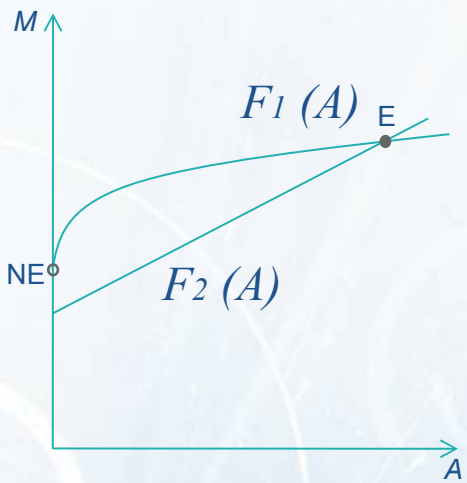
$$NE = \left(0, \frac{\alpha_1}{\lambda_1}\right)$$



Bistable Case

- If α_1 is small, there is only one stationary point, which is globally stable
- For larger values of α_1 there are three stationary points: two extremal points are stable, while the intermediate is unstable

Stability



Monostable Case

$$NE = \left(0, \frac{\alpha_1}{\lambda_1}\right)$$

- If α_1 is high, there are two stationary points, one of them is unstable, and another one is stable

Travelling Waves

Travelling Waves

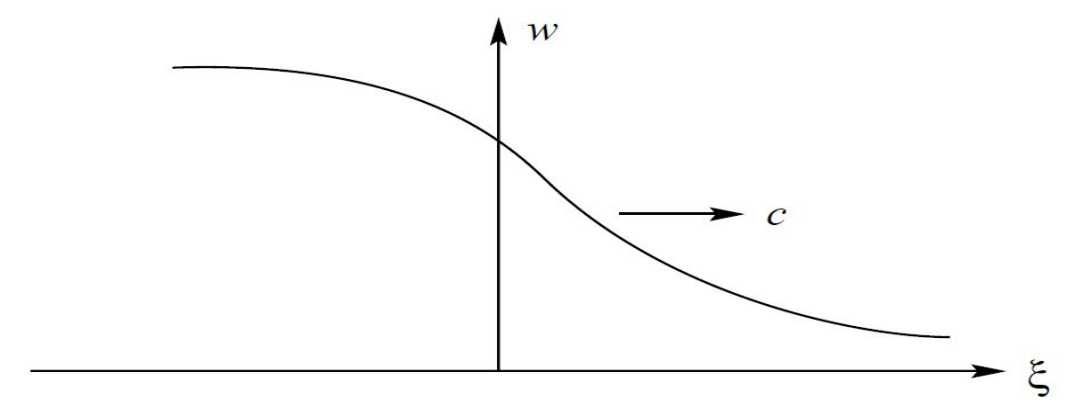
A travelling wave solution of our system is a particular solution representing a front with constant velocity connecting two equilibrium of system , that is a solution of the form:

$$\begin{pmatrix} A(t,x) \\ M(t,x) \end{pmatrix} = W(x-ct)$$

where the constant is a speed of the wave and:

$$\lim_{x \to \pm\infty} W(x) = W_{\pm}$$

with and being two equilibria of system.



$\xi = x - ct$

Existence of solution

Definition: A reaction-diffusion system is called **globally monotone** iff the off-diagonal elements of its **Jacobian** matrix are **positive**

Theorem: In the **bistable case**, there exists a **unique travelling wave solution** connecting the non inflammatory state E_0 and the inflammatory state E_r, i.e. a constant c and a vector valued function $W(x - ct)$, solution of system on the real line and satisfying equation

$$\lim_{x \to \pm\infty} W(x) = W_{\pm}$$

with $W_+ = E_0$ and $W_- = E_r$.

Theorem: In the monostable case, there exists a constant c^* such that there exists a travelling wave solution of velocity c connecting the non inflammatory state and the inflammatory state, i.e. a vector valued function, solution of system on the real line and satisfying equation:

$$\lim_{x \to \pm\infty} W(x) = W_{\pm}$$

with $W_+ = E_0$ and $W_- = E_r$.

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System with diffusion

System

The system with diffusion is a monotone system

Result

Results about existence and stability of travelling waves

Interpretation

Biological interpretation referent to the prediction of the development of the disease

System with diffusion

Point

Low concentrations of cholesterol: there exist only one stationary point

Consequences

Stationary solution that it remains globally stable for the system with diffusion

Interpretation

The atherosclerosis will not develop

System with diffusion

Point

Intermediate cholesterol concentrations: bistable case

Consequences

Travelling wave is unique and globally stable

Interpretation

If its velocity is positive and the initial condition is sufficiently large, then the atherosclerosis will develop

System with diffusion

Point

Large cholesterol Concentrations: monostable case

Consequences

Travelling waves, exist, where c^*, minimal velocity

$\forall c \geq c^*$, where c^* minimal velocity

Interpretation

The inflammation will spread even for small initial perturbations of the disease free equilibrium.

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Aports of the 1-D Model

- Enchainment or the link of cases: in the course of life α_1 increases and so than the organism happens sensitive to inflammations.
- Description like front of propagation: the plaque it stems from a process of propagation amplified conducted by monocytes

Disadvantage of 1-D Model

- No considers that the plaque does not increase infinitely, because there is another wall of the artery, and that implies that we can work with a 2-D model.

2-D Inflamatory Model

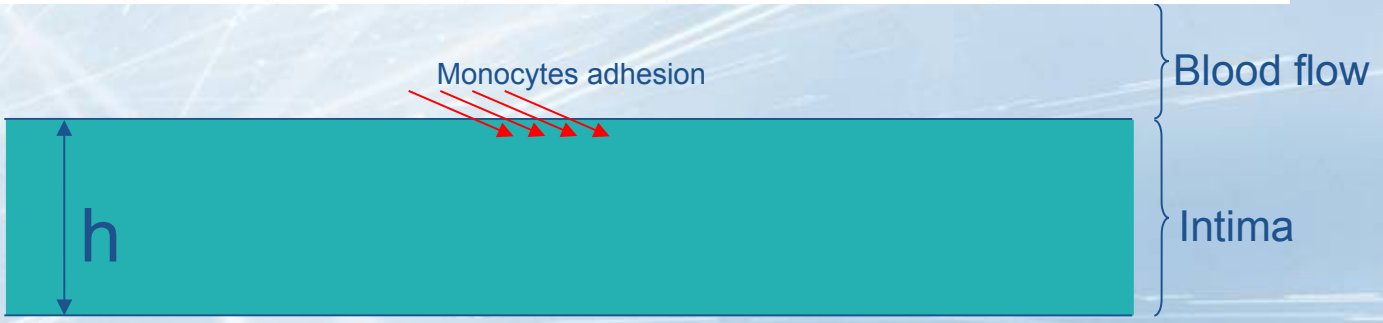
$$\frac{\partial M}{\partial t} = d_M \Delta M - \beta M,$$

$$\frac{\partial A}{\partial t} = d_A \Delta A + f(A)M - \gamma A + b,$$

$$\textit{donde} \quad \Omega = \{(x,y), -\infty < x < \infty,\ 0 \le y \le h\}$$

$$y = 0: \ \frac{\partial M}{\partial y} = 0, \ \frac{\partial A}{\partial y} = 0, \quad y = h: \ \frac{\partial M}{\partial y} = g(A), \ \frac{\partial A}{\partial y} = 0$$

$$M(x,y,0) = M_0(x,y), \quad A(x,y,0) = A_0(x,y).$$



Perspectives

Phase 1	Phase 2	Phase 3
Think about a bound for alpha1	Analyze the existence of the solution for the two-dimensional case	Analyze the stability of these solutions

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Thanks !